

AGI is Coming... Is HCI Ready?

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CCS CONCEPTS

• **Computing methodologies** → **Artificial intelligence**; • **Human-centered computing** → **Human computer interaction (HCI)**.

KEYWORDS

AI; Artificial General Intelligence; AGI; Human-Computer Interaction; HCI; Human-AI Interaction

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1 ABSTRACT

We are at a transformational junction in computing, in the midst of an explosion in capabilities of foundational AI models that may soon match or exceed typical human abilities for a wide variety of cognitive tasks, a milestone often termed Artificial General Intelligence (AGI). Achieving AGI (or even closely approaching it) will transform computing, with ramifications permeating through all aspects of society. This is a critical moment not only for Machine Learning research, but also for the field of Human-Computer Interaction (HCI).

In this talk, I will define what I mean (and what I do NOT mean) by “AGI” (and related concepts, like superintelligence), and my journey from AGI skeptic to believing we are within five to ten years of reaching this milestone. I will then discuss how this new era of computing necessitates a new sociotechnical research agenda on methods and interfaces for studying and interacting with AGI. For instance, how can we extend status quo design and prototyping methods for envisioning novel experiences at the limits of our current imaginations? What novel interaction modalities might AGI (or superintelligence) enable (e.g., “ESP”)? How do we create interfaces for computing systems that may intentionally or unintentionally deceive an end-user? How do we bridge the “gulf of evaluation” when a system may arrive at an answer through methods that fundamentally differ from human mental models, or that may be too complex for an individual user to grasp? How do we evaluate technologies that may have unanticipated systemic side-effects on society when released into the wild?

I will close by reflecting on the relationship between HCI and AI research. Typically, HCI and other sociotechnical domains are not considered as core to the ML research community as areas like model building. However, I argue that research on Human-AI Interaction and the societal impacts of AI is vital and central to this moment in computing history. HCI must not become a “second class citizen” to AI, but rather be recognized as fundamental to ensuring the path to AGI and beyond is a beneficial one.

2 SPEAKER BIO

Meredith Ringel Morris is Director for Human-AI Interaction Research at Google DeepMind. Prior to joining DeepMind, she was Director of the People + AI Research team in Google Research’s Responsible AI division. She also previously served as Research Area Manager for Interaction, Accessibility, and Mixed Reality at Microsoft Research. In addition to her industry role, Dr. Morris has a faculty appointment at the University of Washington, where she is an Affiliate Professor in The Paul G. Allen School of Computer Science & Engineering and also in The Information School. Dr. Morris has been recognized as a Fellow of the ACM and as a member of the ACM SIGCHI Academy for her contributions to Human-Computer Interaction research. She earned her Sc.B. in computer science from Brown University and her M.S. and Ph.D. in computer science from Stanford University. More details on her research and publications are available at <http://merrie.info>.

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